Wavelet Analysis of Sunspot Activity: Exploring Rates of Change and Acceleration

Colucci, D.[1] & Marsellos, A.E.[1]

[1]Dept. Geology, Environment and Sustainability, Hofstra University, NY, USA.

Abstract:

This study presents a detailed time–frequency analysis of daily sunspot numbers spanning 2000–2023, with the aim of characterizing the dynamical behavior of solar activity across multiple temporal scales. Owing to the pronounced nonstationarity and cyclic structure of the raw series, first- and second-order differencing are employed to better isolate underlying temporal dynamics. The resulting first-order (D1) and second-order (D2) series capture the rate of change and the acceleration of sunspot activity, respectively, with stationarity tests confirming their improved suitability for spectral analysis.

Frequency-domain analysis via periodograms indicates that the D1 series retains multiple statistically significant periodic components, with dominant peaks near 2.5, 9.8, and 22.6 days. These periodicities reflect a combination of high-frequency variability and intermediate-scale processes, likely associated with solar rotation and the evolution of active regions. In contrast, the D2 series exhibits a markedly simplified spectral structure, characterized by a dominant short-period peak (~2–3 days), suggesting that second-order differencing effectively suppresses longer-term variability and isolates rapid, transient dynamics.

To further account for the nonstationary nature of solar activity, continuous wavelet transforms are applied to both D1 and D2 series. The wavelet power spectrum of D1 reveals pronounced, time-dependent energy within the 2–30 day band, with intermittent intensification during solar maxima (approximately 2001–2003, 2013–2015, and 2022–2023). This indicates that short-term periodicities evolve in tandem with the solar cycle rather than remaining constant. The D2 wavelet spectrum highlights even higher-frequency variability, with power concentrated below ~15 days and exhibiting localized, fragmented structures associated with abrupt fluctuations in solar activity.

Overall, the comparative analysis demonstrates a clear scale-dependent decomposition of solar dynamics. D1 captures structured short-term variability while preserving intermediate periodicities, whereas D2 isolates high-frequency, transient fluctuations linked to rapid changes in sunspot activity. These findings underscore the inherently nonstationary and multi-scale nature of solar variability and illustrate the effectiveness of combining derivative-based transformations with spectral and wavelet techniques to disentangle the temporal complexity of sunspot behavior.

Background of Study:

- ❏ Solar activity exhibits intrinsic variability driven by complex magnetic processes within the Sun's convection zone.
  The most prominent manifestation of this variability is the Schwabe cycle, an approximately 11-year oscillation in sunspot numbers associated with periodic reversals of the solar magnetic field. While this cycle dominates long-term behavior, it often conceals shorter-term oscillations and secondary modulations that provide critical insight into the Sun's internal dynamics and energy transfer mechanisms.
- ❏ Traditional spectral methods, such as the Fourier transform, are limited when analyzing non-stationary signals containing overlapping frequencies and evolving periodicities. To address these challenges, the Kolmogorov–Zurbenko (KZ) filter offers a flexible, iterative moving-average approach that can effectively isolate specific frequency components from long time series. By applying a KZ filter with a window length exceeding the Schwabe period, long-term trends can be suppressed, enabling detection of high-frequency and transient features in solar data.
- ❏ Monthly sunspot number data from 1818 to 2025 were analyzed to capture both historical and modern variations in solar activity. The KZ filter was used to remove the dominant Schwabe cycle and longer-term components, after which wavelet analysis was applied to the residual series to identify time-varying periodicities and multi-scale modulations. This integrated approach provides a more complete view of solar variability, revealing both persistent and transient cycles that reflect the dynamic nature of the solar magnetic field.

Method/Materials

Data Source:
Daily total sunspot number data (Royal Observatory of Belgium, SILSO World Data Center) (Version 2.0). Data were accessed from the official SILSO repository: http://www.sidc.be/silso/DATA/SN_d_tot_V2.0.csv Dataset: January 1, 2000 through December 31, 2023

Raw daily data were processed and organized in Rstudio using the `data.table`, `dplyr`, and `lubridate` packages to ensure accurate date alignment and time-series integrity.

Data Preprocessing:
First- and second-order differences of the sunspot series were computed. Long-term trends removed, and highlight fluctuations and curvature in solar activity. Stationarity of the original series (also first order d1 and second order d2 series) was evaluated using the Augmented Dickey–Fuller (ADF) test ($p = 0.01$, for d1 and d2 as well).

Spectral Analysis (Periodogram):
Frequency-domain characteristics were examined using a periodogram (`TSA` package) applied to the first-order differenced series. Spectral density was converted to periodicity (1/f) to identify dominant cycles. A nonlinear transformation ($S^5 - S^4$) was applied to suppress noise and enhance peak detection. The strongest periodic component was identified from the maximum spectral density.

Wavelet Transform Analysis:
To examine time-localized periodicities, a Continuous Wavelet Transform (CWT) was performed using the WaveletComp package in R. Wavelet analysis was applied to both first- and second-order differenced datasets to distinguish persistent and transient oscillations.

Parameter settings:
Time step (dt): 1 (daily resolution)
Scale resolution (dj): 1/10
Period range: 2 to 365 days
Mother wavelet: Morlet ($\omega_0 = 6$)
Monte Carlo simulations (n.sim): 10

The wavelet power spectrum was visualized using the `wt.image()` function, displaying normalized power as a function of time and period with quantile-scaled color gradients and 250 contour levels.

1st order Derivative Wavelet

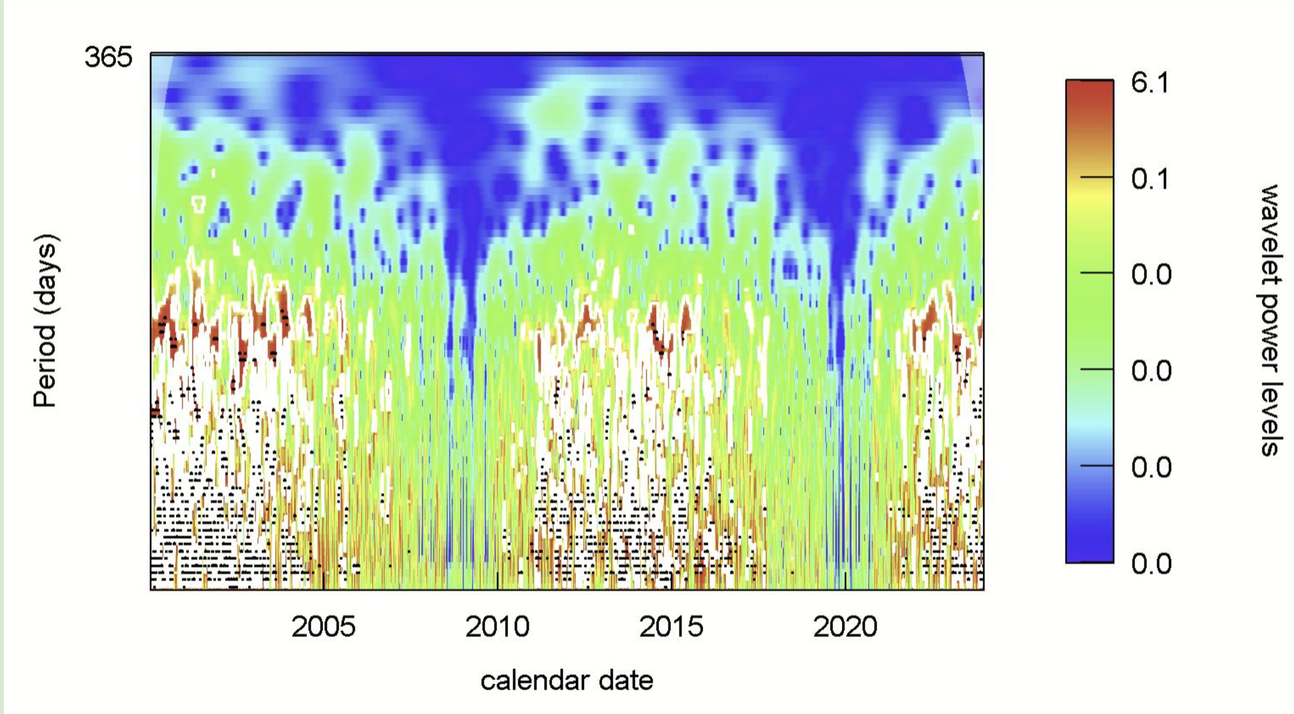

Figure 1 – First-order differenced wavelet spectrum (D1) of sunspot series (2000–2023), showing time-localized periodicities (~2–30 days) with stronger high-frequency power during solar maxima.

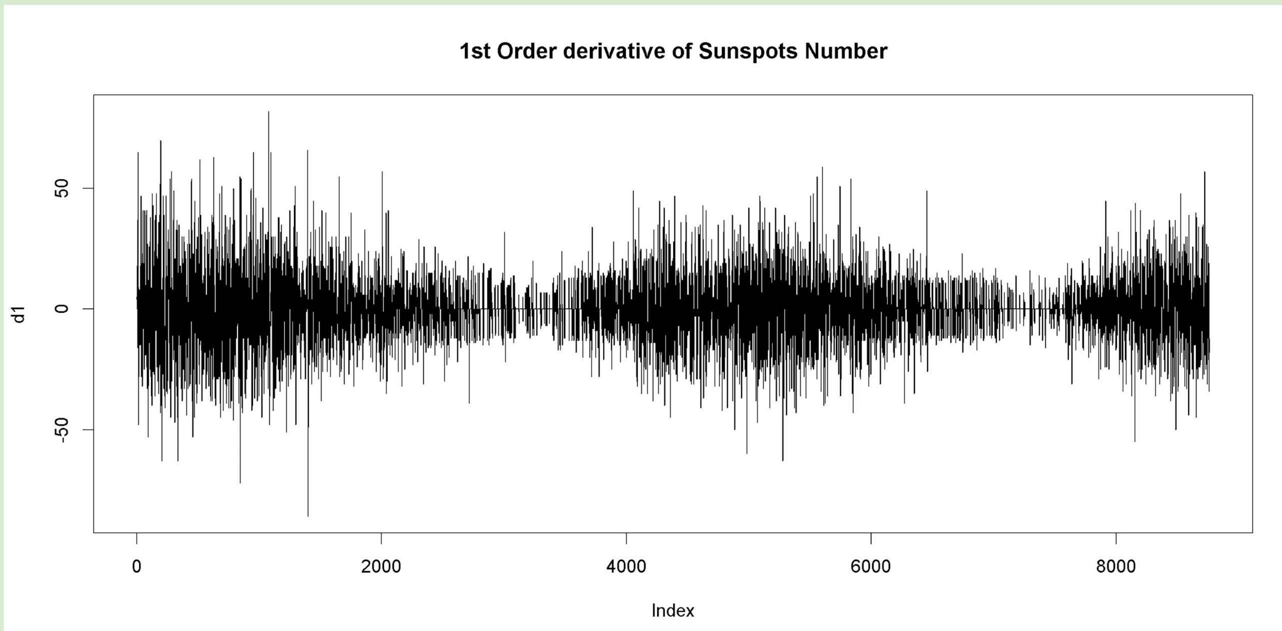

Figure 2 – First-order derivative (D1) time series highlighting short-term variability while retaining intermediate-scale solar cycle structure.

1st order Derivative Results

Time-Series Behavior:
First-order differencing removes long-term trends while preserving the underlying structure of solar variability. The resulting series fluctuates around zero and is stationary (ADF $p = 0.01$), representing the rate of change in sunspot activity.

Spectral Analysis:
The periodogram reveals a dominant low-frequency peak corresponding to the ~11-year solar cycle, consistent with established solar physics.
Application of the nonlinear transformation *($S^5 - S^4$)* enhances peak detection, confirming that this periodicity is a robust feature of the data rather than noise.

Wavelet Analysis:
The wavelet power spectrum of D1 highlights:

- A persistent band of high power associated with the ~11-year cycle
- Temporal modulation of cycle intensity, indicating that solar activity varies in strength across cycles
- Secondary, shorter-period fluctuations that appear intermittently but remain comparatively weak

Interpretation:
D1 captures the dynamics of solar cycle progression, reflecting how sunspot activity increases and decreases over time. The results confirm that while the ~11-year cycle is dominant, it is not constant in amplitude, demonstrating the non-stationary nature of solar variability.

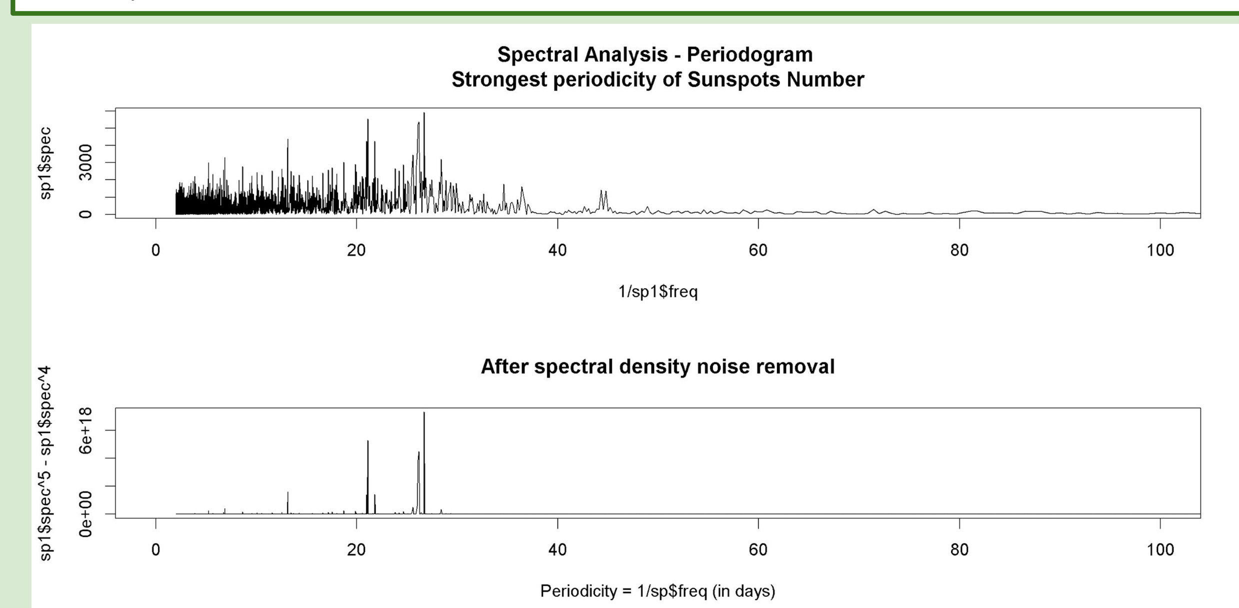

Figure 3 – Periodogram and noise-filtered spectral density Periodogram (top) and noise-suppressed spectrum (bottom) of the D1 series, highlighting dominant periods near ~2.5, ~9.8, and ~22.6 days.

2nd order Derivative Wavelet

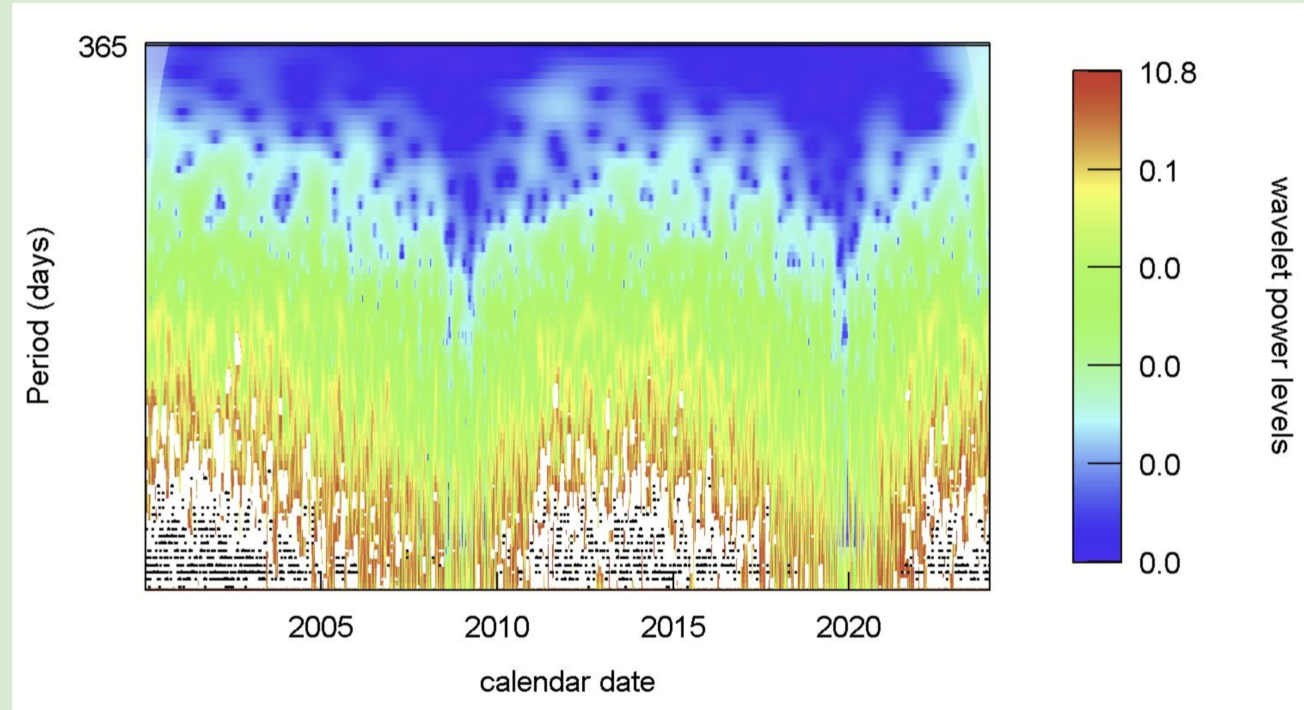

Figure 4 – Second-order differenced wavelet spectrum with power concentrated at shorter periods (<15 days) and more fragmented, transient structures.

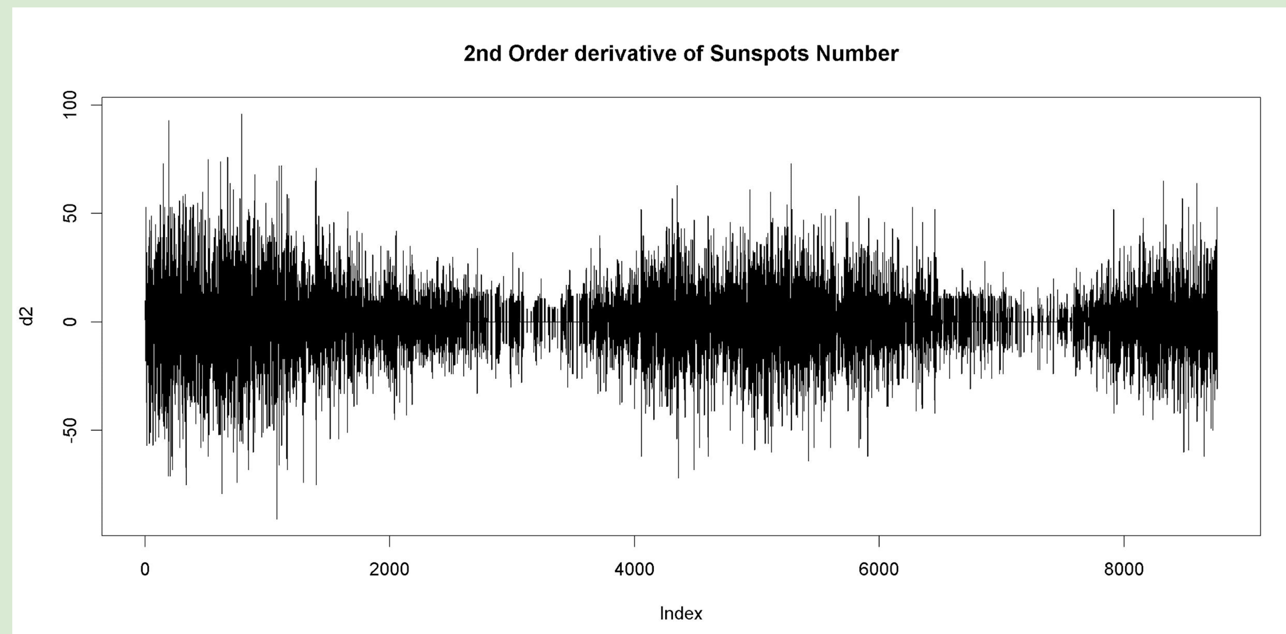

Figure 5 – Second-order derivative (D2) sunspot number time series showing high-frequency variability and rapid fluctuations in solar activity.

2nd order Derivative Results

Time-Series Behavior:
Second-order differencing represents the acceleration (curvature) of sunspot activity—how quickly the rate of change itself is increasing or decreasing. The series is stationary (ADF $p = 0.01$) but exhibits high-frequency oscillations and reduced structural coherence.

Spectral Characteristics:
In contrast to D1:

- The dominant ~11-year cycle is substantially weakened
- Spectral energy shifts toward higher frequencies, reflecting rapid fluctuations
- Periodic structure becomes less distinct

Wavelet Analysis:
The wavelet spectrum for D2 shows:

- Fragmentation of the solar cycle signal, with no consistently strong periodic band
- Increased power at shorter periods, indicating sensitivity to rapid changes in activity
- Reduced temporal coherence, suggesting instability in acceleration patterns

Interpretation:
D2 emphasizes transitions within the solar cycle, such as rapid rises and declines in sunspot activity (e.g., ascent to solar maximum and descent to minimum). However, this transformation diminishes the visibility of the primary cycle and amplifies short-term variability, making long-term periodic structure less interpretable.

Periodogram Analysis Results

The periodogram of the first-order differenced (D1) sunspot series reveals several distinct periodic components in solar variability. When expressed in terms of period (1/frequency), the strongest spectral peaks occur at approximately ~2.5, ~9.8, and ~22.6 days, indicating the presence of dominant short-term cycles.

To improve peak detection, a nonlinear transformation of the spectral density ($S^5 - S^4$) was applied. This effectively suppresses background noise and enhances prominent peaks, making the underlying periodic structure more visible. After noise reduction, the same dominant periodicities remain, confirming their robustness.

The most significant peak corresponds to the strongest periodicity, calculated as the inverse of the frequency with maximum spectral density. This dominant signal is consistent with known solar processes, particularly solar rotation (~26.70623) and shorter-term variations associated with active region dynamics.

Overall, the periodogram results demonstrate that solar activity contains clear and measurable short-term periodic behavior. However, the method does not capture how these periodicities evolve over time, motivating the use of wavelet analysis in subsequent sections.

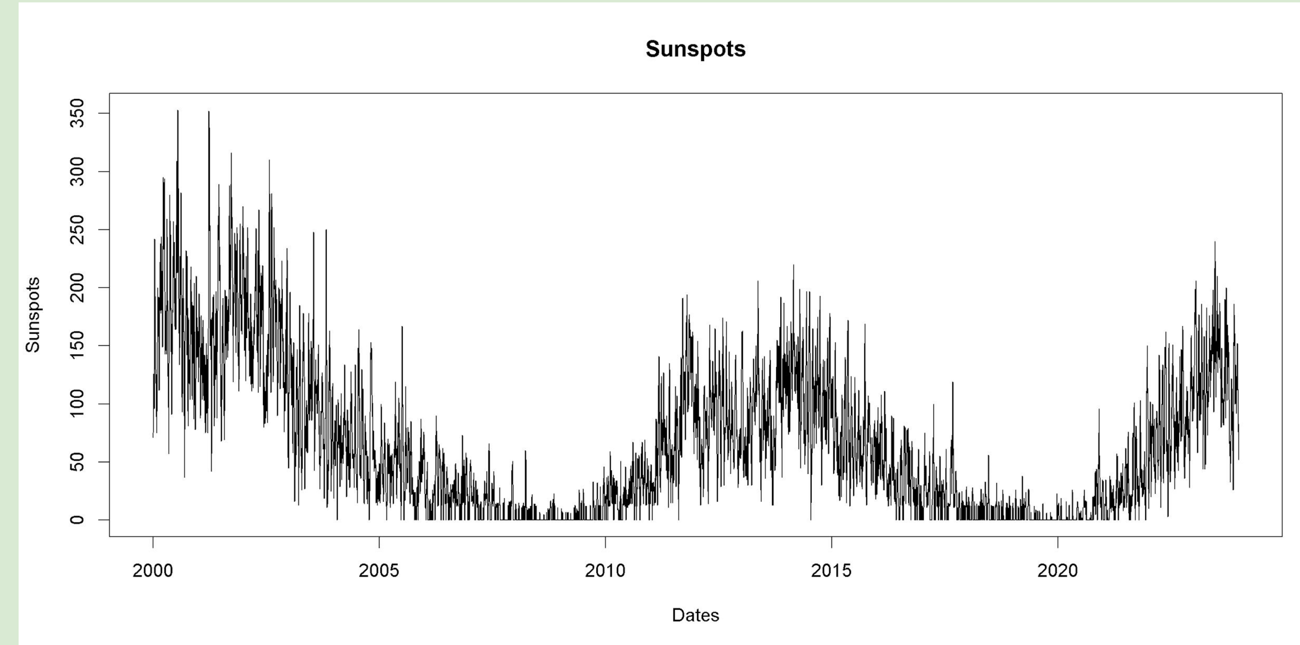

Figure 6 – Raw daily sunspot numbers (2000–2023) showing nonstationary behavior and the characteristic solar cycle pattern.

Discussion

- Solar activity from 2000–2023 exhibits strong nonstationary behavior, with variability spanning multiple time scales. The raw sunspot series reflects the solar cycle, but short-term dynamics are difficult to isolate without transformation. Applying differencing improves stationarity and allows clearer investigation of high-frequency behavior.
- The first-order differenced (D1) series represents the rate of change in solar activity and reveals structured short-term variability. Spectral analysis of D1 identifies dominant periodicities near ~2.5, ~9.8, and ~22.6 days, which are consistent with known solar processes such as solar rotation and the evolution of active regions. After applying nonlinear noise suppression, these peaks become more distinct, indicating that meaningful periodic signals are partially masked by background noise. Wavelet analysis further shows that these periodicities are not constant, but instead vary over time, with stronger and more coherent power during solar maxima (e.g., 2001–2003, 2013–2015, 2022–2023).
- The second-order differenced (D2) series captures the acceleration of solar activity and emphasizes rapid, high-frequency fluctuations. Compared to D1, the D2 signal is more irregular and dominated by short-lived bursts. Wavelet results show power concentrated at shorter periods (<15 days), with fragmented and localized structures. This suggests that D2 isolates transient processes and highlights rapid changes in sunspot dynamics that are less visible in lower-order representations.
- Overall, the results demonstrate that solar variability is driven by both persistent periodic processes and time-dependent fluctuations. The D1 analysis captures stable rotational signals and evolving intermediate-scale structure, while D2 highlights fast, transient behavior. By combining spectral and wavelet methods, this approach provides a more complete understanding of solar dynamics, capturing both global periodic patterns and their evolution over time.

Conclusion:

- This study demonstrates that derivative-based transformations combined with spectral and wavelet analysis provide an effective approach for characterizing the multi-scale and nonstationary dynamics of solar activity.
- The first-order differenced series (D1) successfully captures structured periodic variability, including short- to intermediate-scale oscillations associated with solar rotation and active region evolution, while preserving sensitivity to modulation across the solar cycle. In contrast, the second-order series (D2) isolates high-frequency, transient fluctuations, reflecting rapid changes in sunspot activity but reducing the visibility of sustained periodic behavior.
- Together, these results highlight a clear scale-dependent decomposition of solar variability:
- D1 represents organized, physically meaningful dynamics
- D2 reflects short-lived, high-frequency processes
- Overall, the findings confirm that solar activity is inherently nonstationary, driven by the interaction of cyclical patterns and transient events, and demonstrate the value of combining differencing with time–frequency methods to better understand solar behavior.

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Credit Authorship Contribution Statement:

Dominick Colucci: R coding, figures, poster, writing all sections.